Heer Rana

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Education

Adani University 2023 – Present

B.Tech in Computer Science and Engineering (AI-ML) | CGPA: 8.19 *Ahmedabad, Gujarat*

- **Relevant Coursework:** Data Structures & Algorithms, OOPS, DBMS, Machine Learning, Deep Learning, IOT & Automation, Computer Networks, Operating Systems.

Experience

Adani University Jun 2026 – Jul 2026

Research Intern (Skin Cancer Detection Using Vision Transformer) *Ahmedabad, Gujarat*

- Engineered a robust deep learning pipeline under Dr. Abhay Sharma for 7-class skin lesion classification utilizing the HAM10000 dataset comprising 10,015 dermoscopic images.
- Developed an automated image preprocessing workflow by implementing Gaussian filtering and CLAHE contrast enhancement, reducing visual artifacts across 100% of the training data.
- Executed 4 progressive experimental phases incorporating Focal Loss, MixUp augmentation, and early stopping, accelerating model convergence time by 20%.
- Evaluated and benchmarked ResNet50 against ViT-B/16 architectures, where ViT-B/16 achieved a superior peak accuracy of 85.55%, outperforming ResNet50's 80.69%.

Projects

Snippy | *Code Sharing SaaS* | *Next.js, TypeScript, Gemini AI, MongoDB* | GitHub Oct 2026 – Present

- Architected an intelligent developer SaaS supporting secure code sharing, complete with advanced syntax highlighting and automatic formatting for over 50 programming languages.
- Integrated Google Generative AI to provide automated code quality scoring and generate documentation, reducing manual code review time by 40%.
- Built real-time community features including live status indicators and instant QR-code sharing, achieving sub-50ms synchronization latency utilizing Tailwind CSS and MongoDB.

TeamSync | *Collaboration Platform* | *MERN, TypeScript, Socket.io, Tailwind* | GitHub Aug 2026

- Constructed a full-stack project management platform enabling granular role-based access control (Admin, Manager, Member, Viewer) across customizable task boards, improving team workflow efficiency by 30%.
- Implemented real-time collaboration capabilities utilizing Socket.io to support up to 100 concurrent users with live task updates and activity notifications without page reloads.
- Secured platform access using Google OAuth and JWT, delivering 3 comprehensive analytics dashboards utilizing React and Node.js to track key performance metrics.

IMDb Sentiment Analysis | *Python, TensorFlow, Keras, LSTM* | GitHub Apr 2026

- Designed an end-to-end sentiment analysis model leveraging a Bidirectional LSTM network on 50,000 IMDb reviews, securing an 83.5% test accuracy.
- Optimized text data preprocessing pipelines via TextVectorization, establishing a 20,000-word vocabulary and 250-token sequence length for efficient computational modeling.
- Deployed a real-time prediction script integrating the trained model and vocabulary, accelerating text inference speeds to process user inputs in under 200 milliseconds.

Technical Skills

Languages: JavaScript, TypeScript, Python, C, Java, HTML, CSS
Web Technologies: React 18, Next.js 14, Node.js, Express.js, Tailwind CSS, Socket.IO, REST APIs
Databases & Cloud: MongoDB, SQL, Appwrite, Cloudinary, Vercel, Render
AI & ML: Google Generative AI, TensorFlow, Keras, Pandas, NumPy, Scikit-learn
Tools & Architecture: Git, GitHub, VS Code, Figma, Postman, JWT Authentication, MVC

Certifications & Achievements

- **Machine Learning Foundations** – AWS Academy Graduate (Nov 2025)
- **LLMs using Python (2-Day Workshop)** – Techfest, IIT Bombay (Dec 2025)
- **Prompt Engineering** – Cognitive Class, IBM (Dec 2025)